

The synaptic Package

User Guide and Feature Showcase

Template Development Team

CTAN Community

E-mail: synaptic@example.com

Abstract: This document serves as a hands-on demonstration of `synaptic`, a modern academic typesetting framework built on `expl3` and `KOMA-Script`. All major features – language switching, colour themes, theorem environments, extra math alphabets, table of contents layouts, and micro-typographic fine-tuning – are used here as live examples. Compilation requires `LuaLaTeX`.

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To the elegance of structured typesetting.

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1 Quick Start

Load the package with your desired options. This document uses `\color=carbon`, `\toc-layout=inline`, and `\extra-math=true`:

```
\usepackage[
  lang=en, color=carbon, style=modern,
  extra-math=true, toc=true,
  toc-layout=inline, fine-breaking=false
]{synaptic}
```

2 Title Page Customization

The following commands set metadata on the title page:

```
\title{...}
\author[mark]{...}
\affiliation[mark]{...}
\emailAdd{...}
\keywords{...}
\fpheader{...}
\subheader{...}
\preprint{...}
\arxivnumber{...}
\dedicated{...}
```

The abstract is entered as an environment, supporting multiple paragraphs.

3 Theorem Environments

`synaptic` provides `theorem`, `lemma`, `proposition`, and `definition` by default. Their names are automatically localised when `lang=zh` is used.

You can extend the set with `\declarecustomtheorem`:

```
\declarecustomtheorem{example}{Example}
\declarecustomtheorem{remark}{Remark}
```

Definition 3.1 (Linear Space). *Let V be a vector space over a field $\mathbb{F}$. If $\dim V = n$, then $V \cong \mathbb{F}^n$.*

Theorem 3.2 (Dimension Formula). *For any linear transformation $T : V \rightarrow W$,*

$$\dim \ker T + \dim \operatorname{im} T = \dim V. \quad (3.1)$$

Lemma 3.3. *If T is invertible, then $\dim V = \dim W$.*

Proof. Immediate from Theorem 3.2. ■

Example 3.4 (Matrix Representation). *In a chosen basis, T corresponds to a matrix $A \in \mathrm{GL}_n(\mathbb{R})$.*

Remark 3.5. *Custom environments share the same counter as theorem.*

4 Extra Math Alphabets

When `\extra-math=true`, the following shortcuts are available:

- `\symbb{Q}` → $\mathbb{Q}$ (blackboard bold)
- `\symcal{C}` → $\mathcal{C}$ (calligraphic)
- `\symfrak{g}` → $\mathfrak{g}$ (Fraktur)

They integrate seamlessly with `unicode-math`.

5 Table of Contents and Layouts

The `\toc-layout` option controls where the table of contents appears:

- `top` – after the title page (default)
- `inline` – inside the title page (as demonstrated here)
- `page` – on a separate page

The `\style` option provides two page geometries:

- `modern` – standard margins for comfortable reading
- `minimal` – compact layout for short notes

6 Fine-Breaking and Micro-typography

When `\fine-breaking=true`, the package adjusts `\emergencystretch`, `\tolerance`, and penalties to avoid widows and overfull boxes. Here it is turned off (`\fine-breaking=false`) to show the difference.

7 Colour Themes

Three preset colour themes are available:

- ocean – deep blue (default)
- teal – teal green
- carbon – dark grey (used in this document)

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